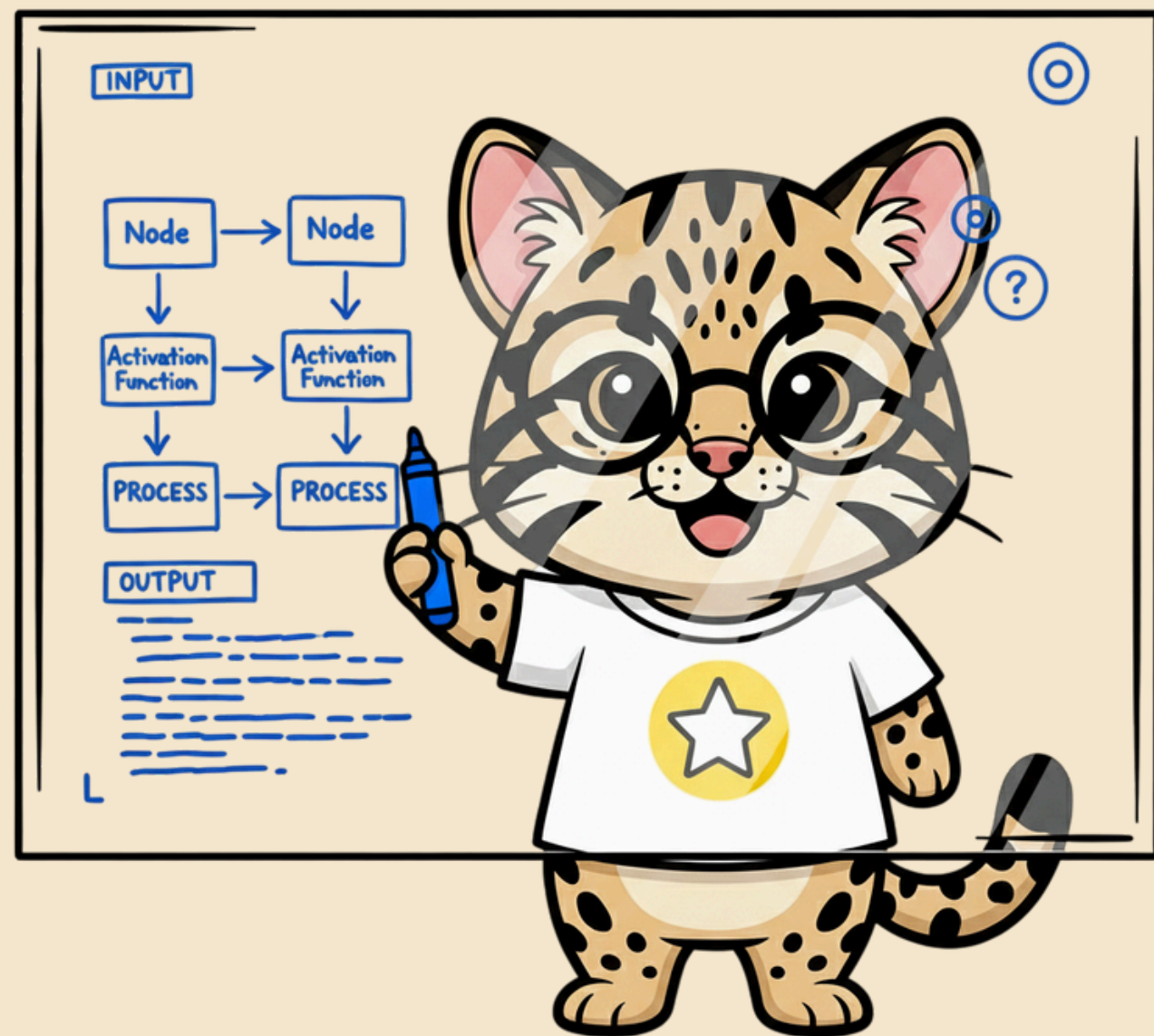
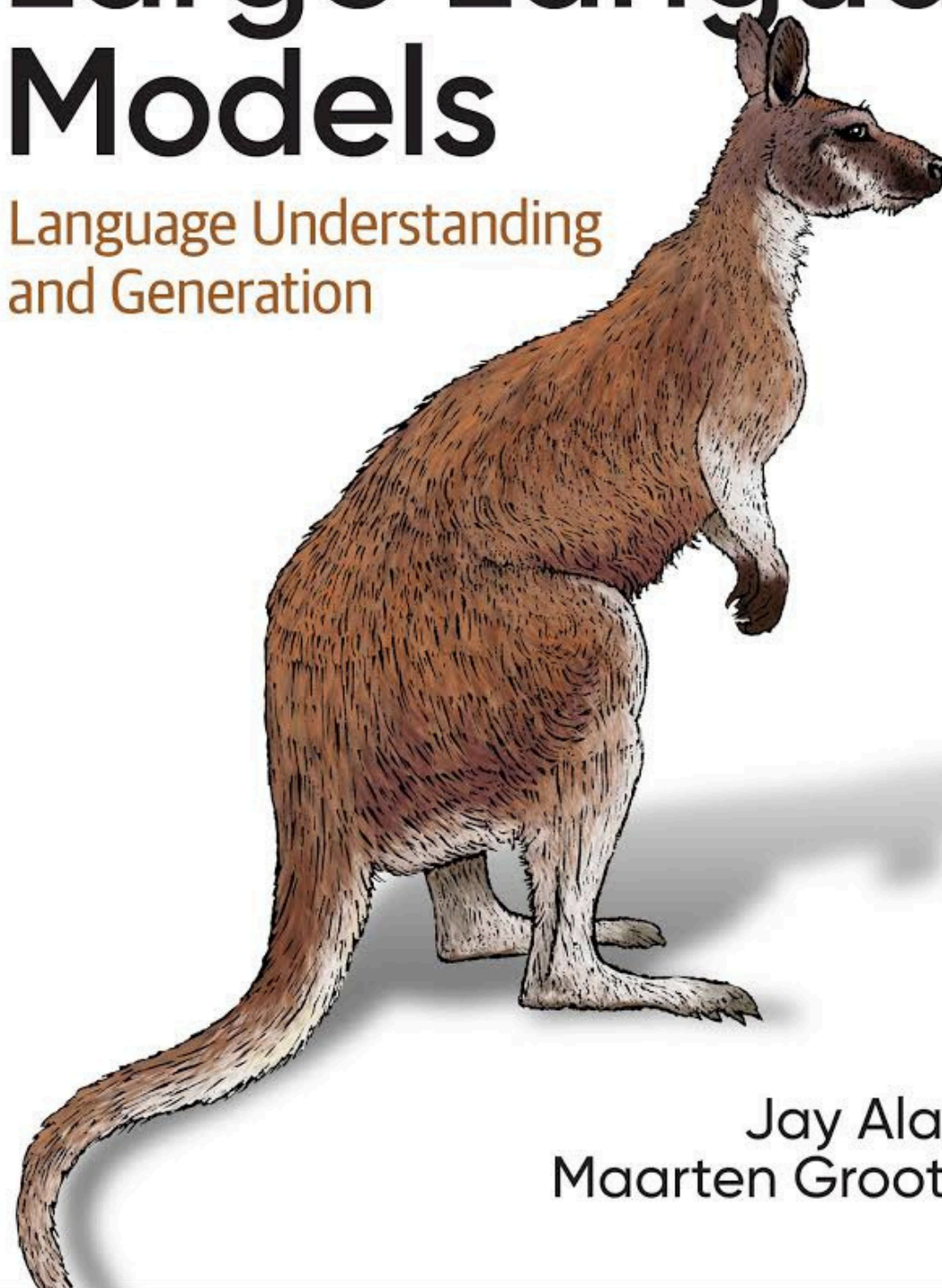




O'REILLY®

Hands-On Large Language Models

Language Understanding
and Generation

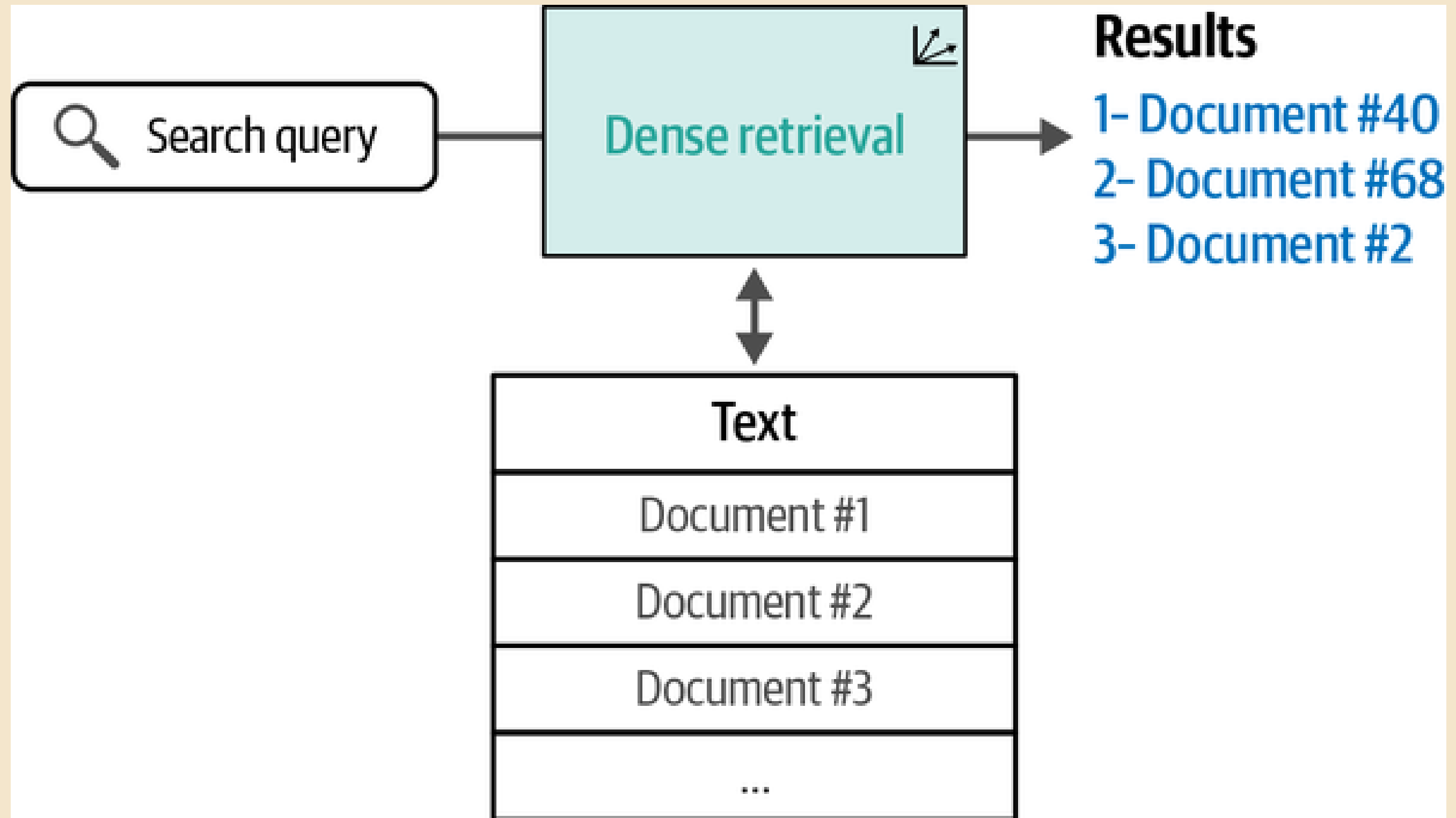


Jay Alammar &
Maarten Grootendorst

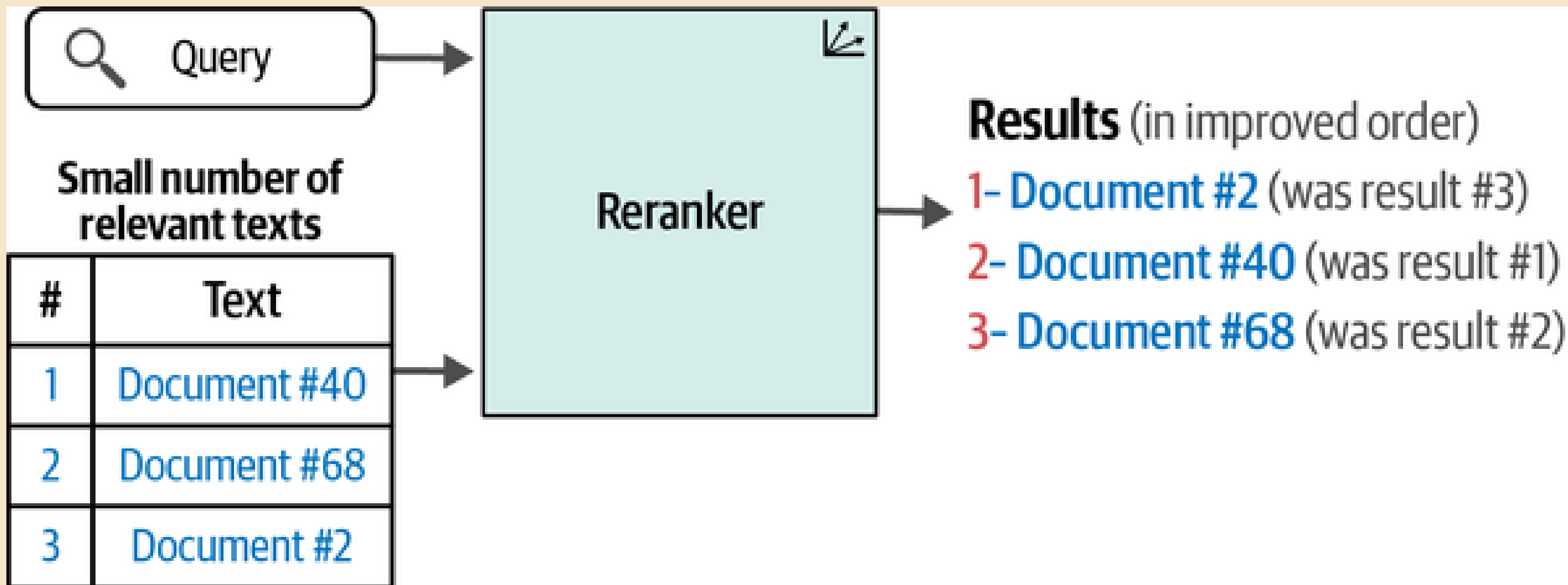
Overview of Semantic Search and RAG

- 密集檢索 (Dense Retrieval)
- 重新排序 (Reranking)
- RAG (檢索增強生成)

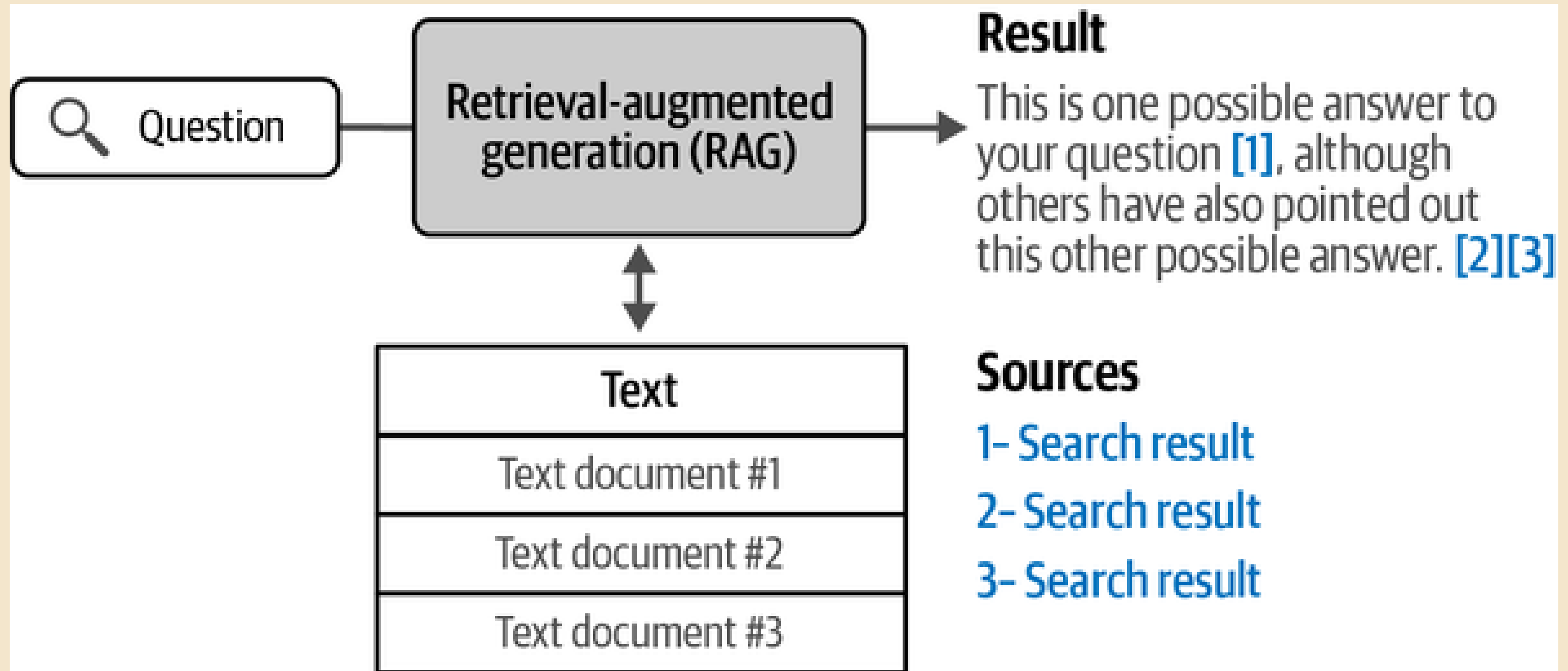




密集検索 (Dense Retrieval)

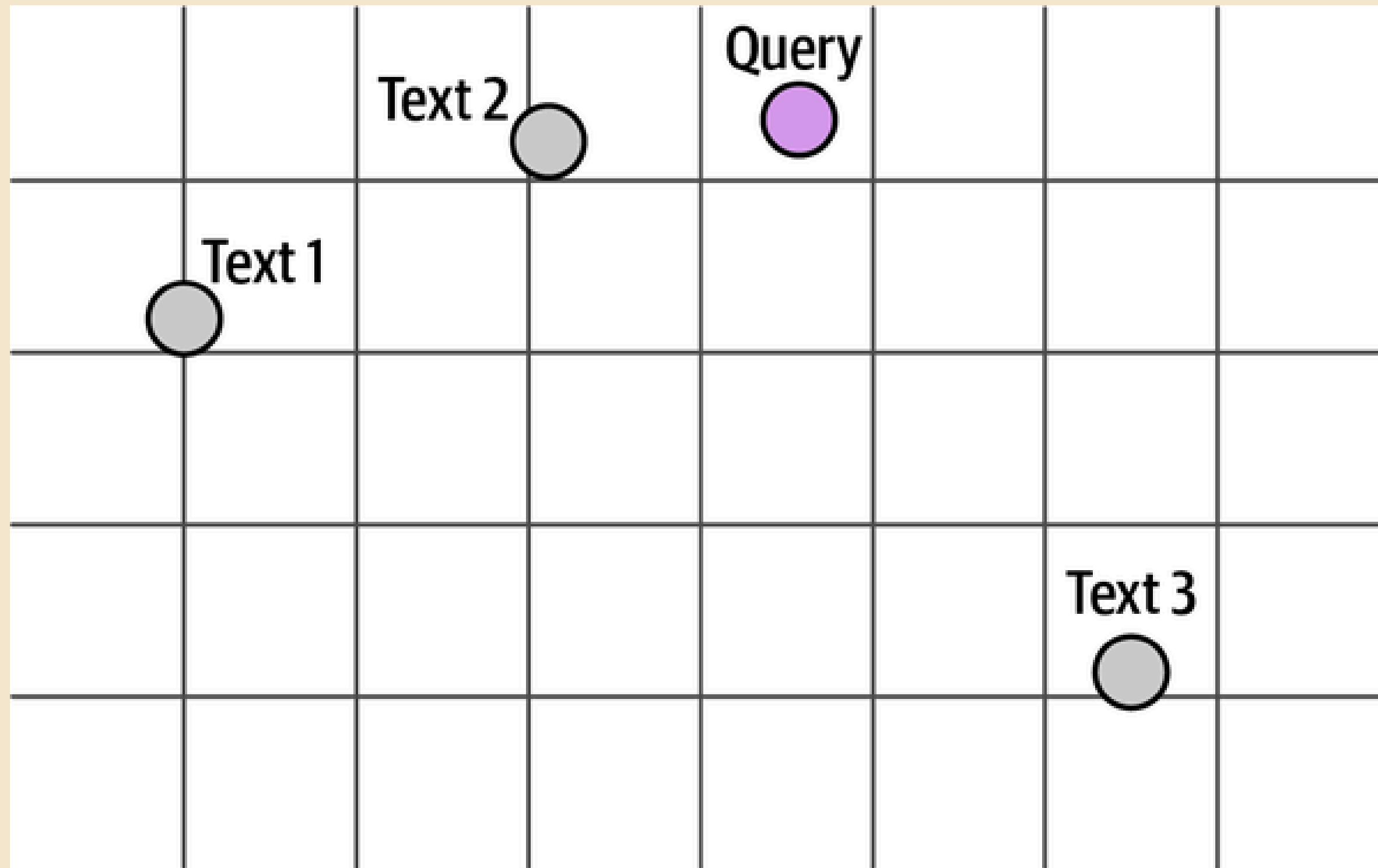


重新排序 (Reranking)

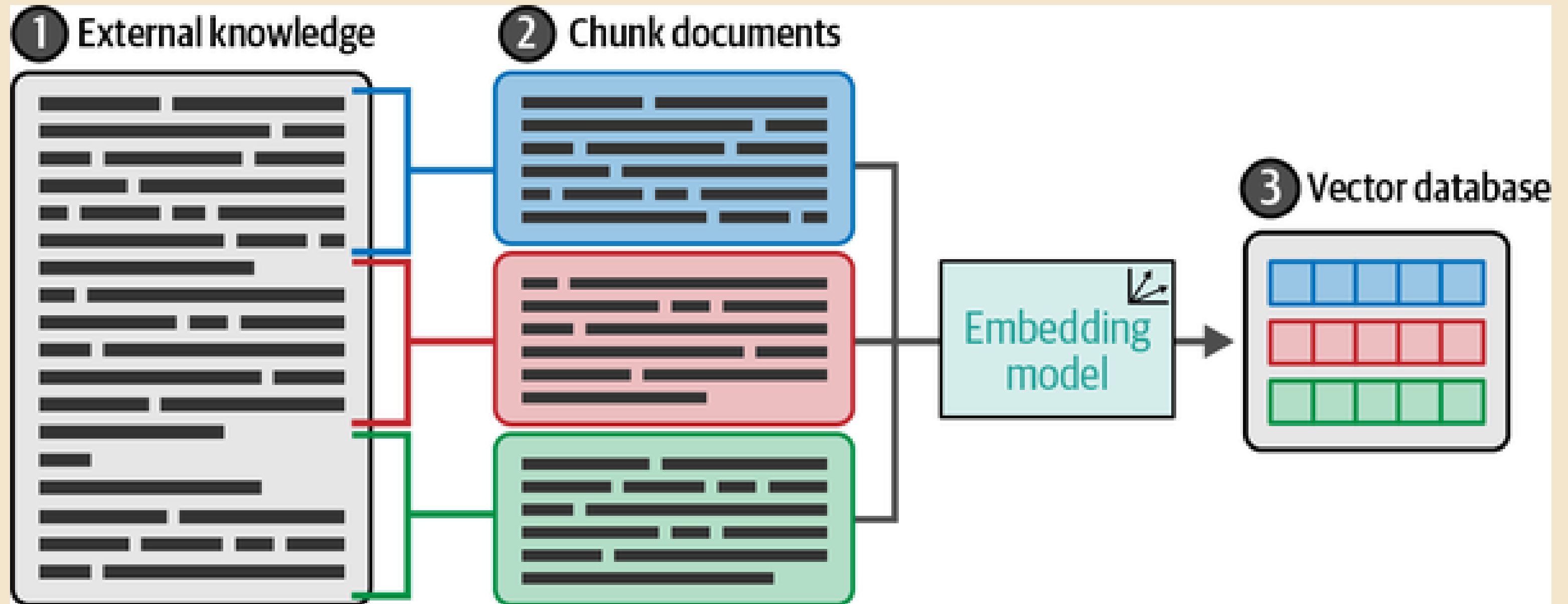


RAG（検索増強生成）

密集檢索 (Dense Retrieval)



文件切分



One vector per document

Interstellar (film) 15, 88 languages

Article **Tab** Read Edit View history Tools

From Wikipedia, the free encyclopedia

Interstellar is a 2014 epic science fiction drama film directed by Christopher Nolan, who cowrote it with his brother Jonathan. It stars Matthew McConaughey, Anne Hathaway, Jessica Chastain, Bill Irwin, Ellen Burstyn, Michael Caine, and Matt Damon. Set in a dystopian future where Earth is suffering from catastrophic drought and famine, the film follows a group of astronauts who travel through a wormhole near Saturn in search of a new home for humankind.

The screenplay had its origins in a script Jonathan developed in 2007, and was originally set to be directed by Steven Spielberg. Theoretical physicist Kip Thorne was an executive producer and scientific consultant on the film, and wrote the film's book *The Science of Interstellar*. Cinematographer Wally Pfister shot it on 35-mm movie film in the Periscope anamorphic format and IMAX 70 mm. Filming began in late 2012 and took place in Alberta, Klausur, and Los Angeles. Interstellar uses extensive practical and miniature effects, and the company DNEG created additional digital effects.

Interstellar premiered in Los Angeles on October 26, 2014. In the United States, it was not released on **film stock**, expanding to venues using digital projections. The film received positive reviews from critics and grossed over \$687 million worldwide (\$708 million after subsequent re-releases), making it the **fourth-highest-grossing film of 2014**. Thorne's **computer-generated** depiction of a **black hole** in the film has also received commendation from astronomers and physicists. **PFFP** Interstellar was nominated for five awards at the **87th Academy Awards**, winning **Best Visual Effects**, and received numerous other accolades.

Interstellar

Theatrical release poster

Directed by	Christopher Nolan
Written by	Jonathan Nolan Christopher Nolan
Produced by	Emma Thomas Christopher Nolan Lynda Olin
Starring	Matthew McConaughey Anne Hathaway Jessica Chastain

Chunk document into multiple chunks

Interstellar (film) 15, 88 languages

Article **Tab** Read Edit View history Tools

From Wikipedia, the free encyclopedia

Interstellar is a 2014 epic science fiction drama film directed by Christopher Nolan, who cowrote it with his brother Jonathan. It stars Matthew McConaughey, Anne Hathaway, Jessica Chastain, Bill Irwin, Ellen Burstyn, Michael Caine, and Matt Damon. Set in a dystopian future where Earth is suffering from catastrophic drought and famine, the film follows a group of astronauts who travel through a wormhole near Saturn in search of a new home for humankind.

The screenplay had its origins in a script Jonathan developed in 2007, and was originally set to be directed by Steven Spielberg. Theoretical physicist Kip Thorne was an executive producer and scientific consultant on the film, and wrote the film's book *The Science of Interstellar*. Cinematographer Wally Pfister shot it on 35-mm movie film in the Periscope anamorphic format and IMAX 70 mm. Filming began in late 2012 and took place in Alberta, Klausur, and Los Angeles. Interstellar uses extensive practical and miniature effects, and the company DNEG created additional digital effects.

Interstellar premiered in Los Angeles on October 26, 2014. In the United States, it was not released on **film stock**, expanding to venues using digital projections. The film received positive reviews from critics and grossed over \$687 million worldwide (\$708 million after subsequent re-releases), making it the **fourth-highest-grossing film of 2014**. Thorne's **computer-generated** depiction of a **black hole** in the film has also received commendation from astronomers and physicists. **PFFP** Interstellar was nominated for five awards at the **87th Academy Awards**, winning **Best Visual Effects**, and received numerous other accolades.

Interstellar

Theatrical release poster

Directed by	Christopher Nolan
Written by	Jonathan Nolan Christopher Nolan
Produced by	Emma Thomas Christopher Nolan Lynda Olin
Starring	Matthew McConaughey Anne Hathaway Jessica Chastain

 Document vector

 Chunk 1 vector

 Chunk 2 vector

 Chunk 3 vector

Each sentence is a chunk

Interstellar (film)

15, 48 paragraphs

Article · Film · 2015

From Wikipedia, the free encyclopedia

Interstellar is a 2014 epic science fiction drama film directed by Christopher Nolan, who cowrote it with his brother Jonathan. It stars Matthew McConaughey, Anne Hathaway, Jessica Chastain, Bill Irwin, Ellen Burstyn, Michael Caine, and Matt Damon. Set in a dystopian future where Earth is suffering from catastrophic drought and famine, the film follows a group of astronauts who travel through a wormhole near Saturn in search of a new home for humanity.

The screenplay had its origins in a script Jonathan developed in 2007, and was originally set to be directed by Steven Spielberg. Theoretical physicist Kip Thorne was an executive producer and scientific consultant on the film, and wrote the book *The Science of Interstellar*. Cinematographer Wally Pfister highlights that it is the most made film in the Panavision anamorphic format and IMAX 70 mm. Filming began in late 2013 and took place in Atlanta, Houston, and Los Angeles. Interstellar uses extensive practical and miniature effects, and the company IMAX created additional digital effects.

Interstellar premiered in Los Angeles on October 26, 2014. In the United States, it was first released on *film stock*, expanding to venues using digital projections. The film received positive reviews from critics and grossed over \$670 million worldwide (\$700 million after subsequent re-releases), making it the *third-highest-grossing film of 2014*. Thorne's computer-generated depiction of a black hole in the film has also received commendation from astronomers and physicists. *Interstellar* was nominated for five awards at the 87th Academy Awards, winning Best Visual Effects, and received numerous other accolades.

Thumbnail image poster

Directed by

Written by

Produced by

Starring

Christopher Nolan

Jonathan Nolan

Christopher Nolan

James Thorne

Christopher Nolan

Lynzee Kling

Matthew McConaughey

Anne Hathaway

Jessica Chastain

Each paragraph is a chunk

Interstellar (film)

15, 48 paragraphs

Article · Film · 2015

From Wikipedia, the free encyclopedia

Interstellar is a 2014 epic science fiction drama film directed by Christopher Nolan, who cowrote it with his brother Jonathan. It stars Matthew McConaughey, Anne Hathaway, Jessica Chastain, Bill Irwin, Ellen Burstyn, Michael Caine, and Matt Damon. Set in a dystopian future where Earth is suffering from catastrophic drought and famine, the film follows a group of astronauts who travel through a wormhole near Saturn in search of a new home for humanity.

The screenplay had its origins in a script Jonathan developed in 2007, and was originally set to be directed by Steven Spielberg. Theoretical physicist Kip Thorne was an executive producer and scientific consultant on the film, and wrote the book *The Science of Interstellar*. Cinematographer Wally Pfister highlights that it is the most made film in the Panavision anamorphic format and IMAX 70 mm. Filming began in late 2013 and took place in Atlanta, Houston, and Los Angeles. Interstellar uses extensive practical and miniature effects, and the company IMAX created additional digital effects.

Interstellar premiered in Los Angeles on October 26, 2014. In the United States, it was first released on *film stock*, expanding to venues using digital projections. The film received positive reviews from critics and grossed over \$670 million worldwide (\$700 million after subsequent re-releases), making it the *third-highest-grossing film of 2014*. Thorne's computer-generated depiction of a black hole in the film has also received commendation from astronomers and physicists. *Interstellar* was nominated for five awards at the 87th Academy Awards, winning Best Visual Effects, and received numerous other accolades.

Thumbnail image poster

Directed by

Written by

Produced by

Starring

Christopher Nolan

Jonathan Nolan

Christopher Nolan

James Thorne

Christopher Nolan

Lynzee Kling

Matthew McConaughey

Anne Hathaway

Jessica Chastain

Overlapping window of sentences

Interstellar (film)

15, 48 paragraphs

Article · Film · 2015

From Wikipedia, the free encyclopedia

Interstellar is a 2014 epic science fiction drama film directed by Christopher Nolan, who cowrote it with his brother Jonathan. It stars Matthew McConaughey, Anne Hathaway, Jessica Chastain, Bill Irwin, Ellen Burstyn, Michael Caine, and Matt Damon. Set in a dystopian future where Earth is suffering from catastrophic drought and famine, the film follows a group of astronauts who travel through a wormhole near Saturn in search of a new home for humanity.

The screenplay had its origins in a script Jonathan developed in 2007, and was originally set to be directed by Steven Spielberg. Theoretical physicist Kip Thorne was an executive producer and scientific consultant on the film, and wrote the book *The Science of Interstellar*. Cinematographer Wally Pfister highlights that it is the most made film in the Panavision anamorphic format and IMAX 70 mm. Filming began in late 2013 and took place in Atlanta, Houston, and Los Angeles. Interstellar uses extensive practical and miniature effects, and the company IMAX created additional digital effects.

Interstellar premiered in Los Angeles on October 26, 2014. In the United States, it was first released on *film stock*, expanding to venues using digital projections. The film received positive reviews from critics and grossed over \$670 million worldwide (\$700 million after subsequent re-releases), making it the *third-highest-grossing film of 2014*. Thorne's computer-generated depiction of a black hole in the film has also received commendation from astronomers and physicists. *Interstellar* was nominated for five awards at the 87th Academy Awards, winning Best Visual Effects, and received numerous other accolades.

Thumbnail image poster

Directed by

Written by

Produced by

Starring

Christopher Nolan

Jonathan Nolan

Christopher Nolan

James Thorne

Christopher Nolan

Lynzee Kling

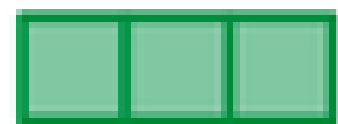
Matthew McConaughey

Anne Hathaway

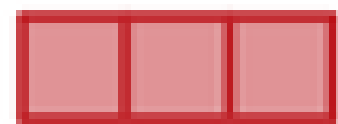
Jessica Chastain



Chunk 1 vector

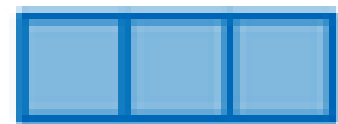


Chunk 2 vector



Chunk 3 vector

• • •



Chunk 15 vector



Chunk 1 vector



Chunk 2 vector



Chunk 3 vector



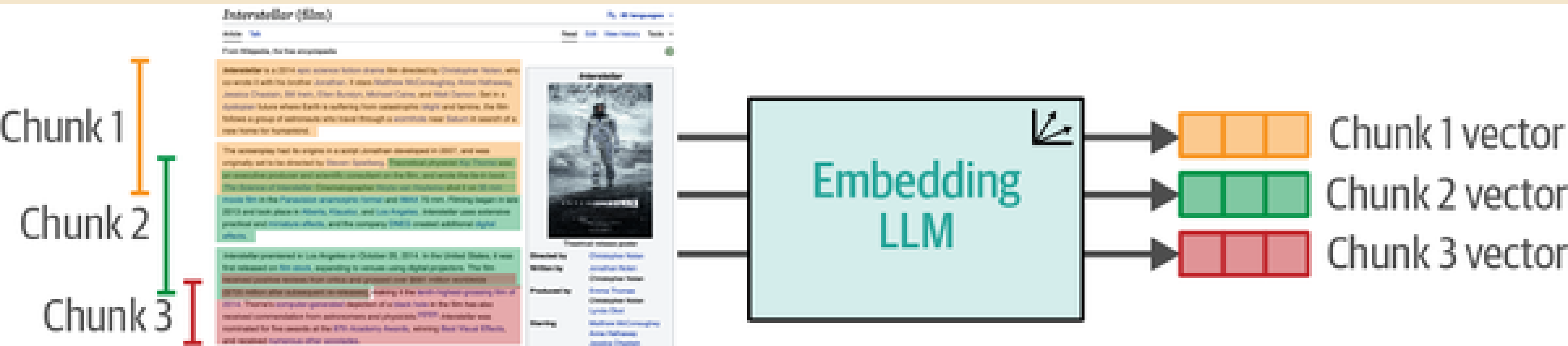
Chunk 1 vector



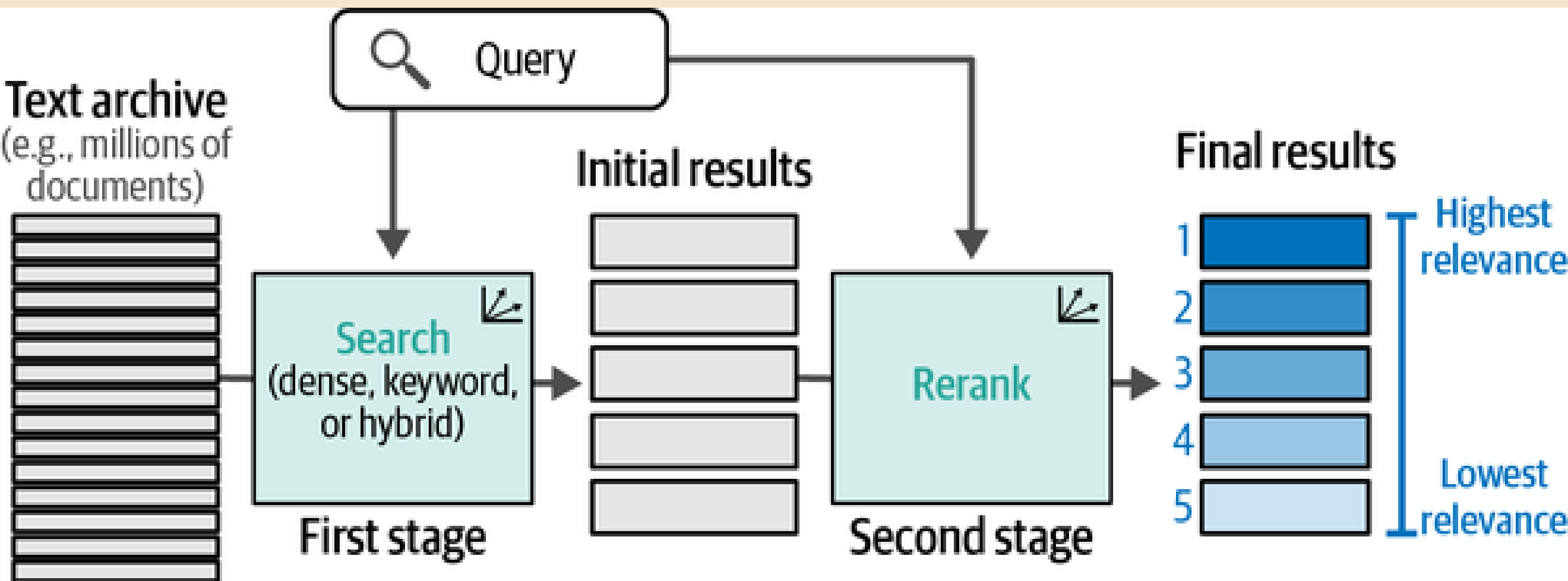
Chunk 2 vector

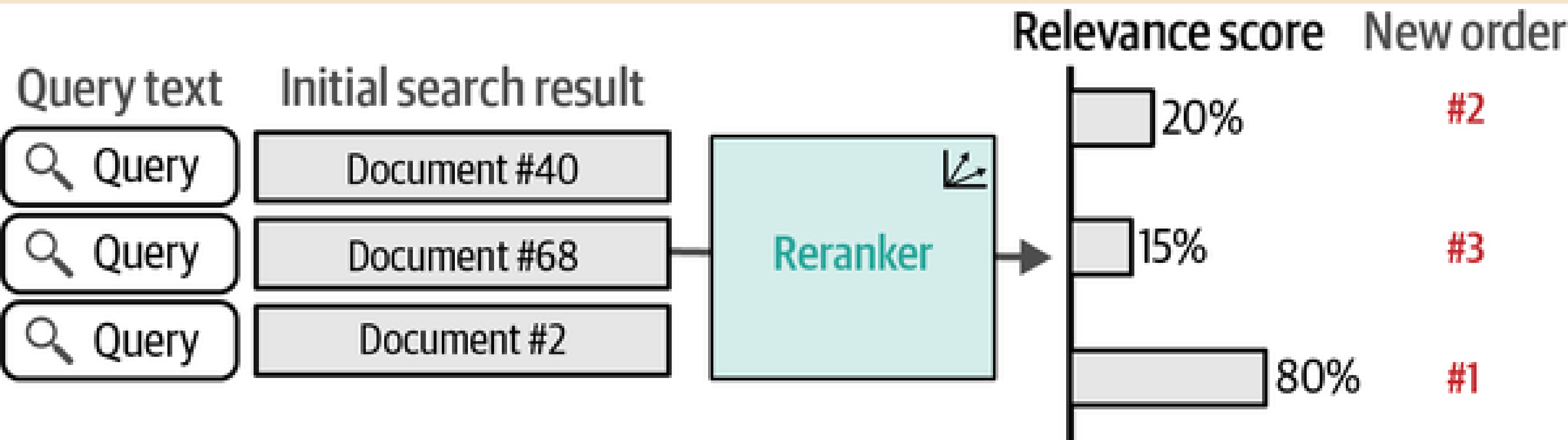


Chunk 3 vector



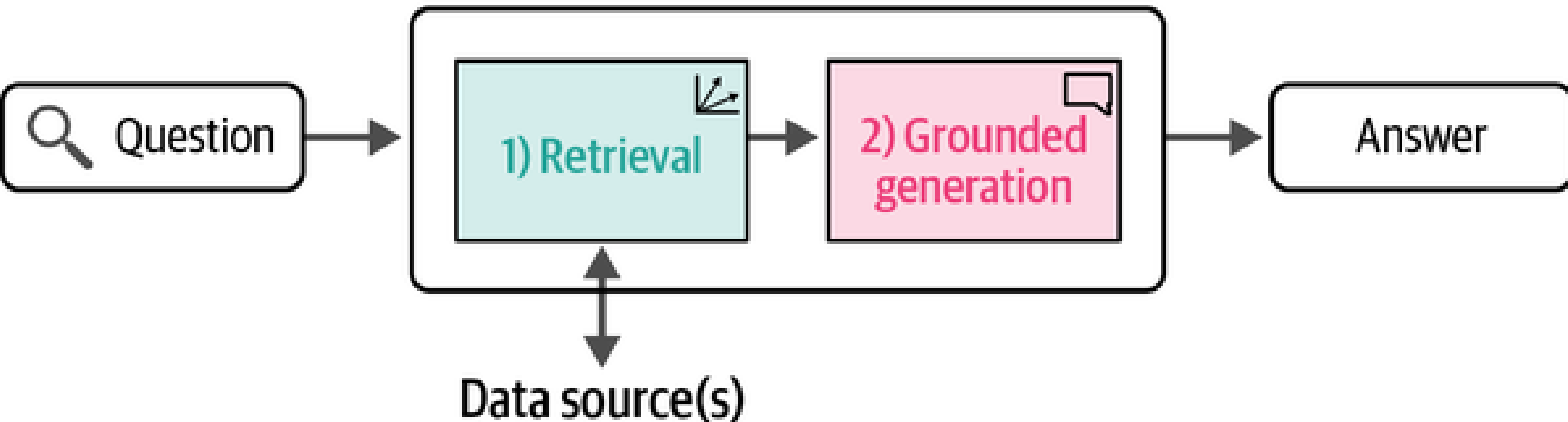
重新排序 (Reranking)

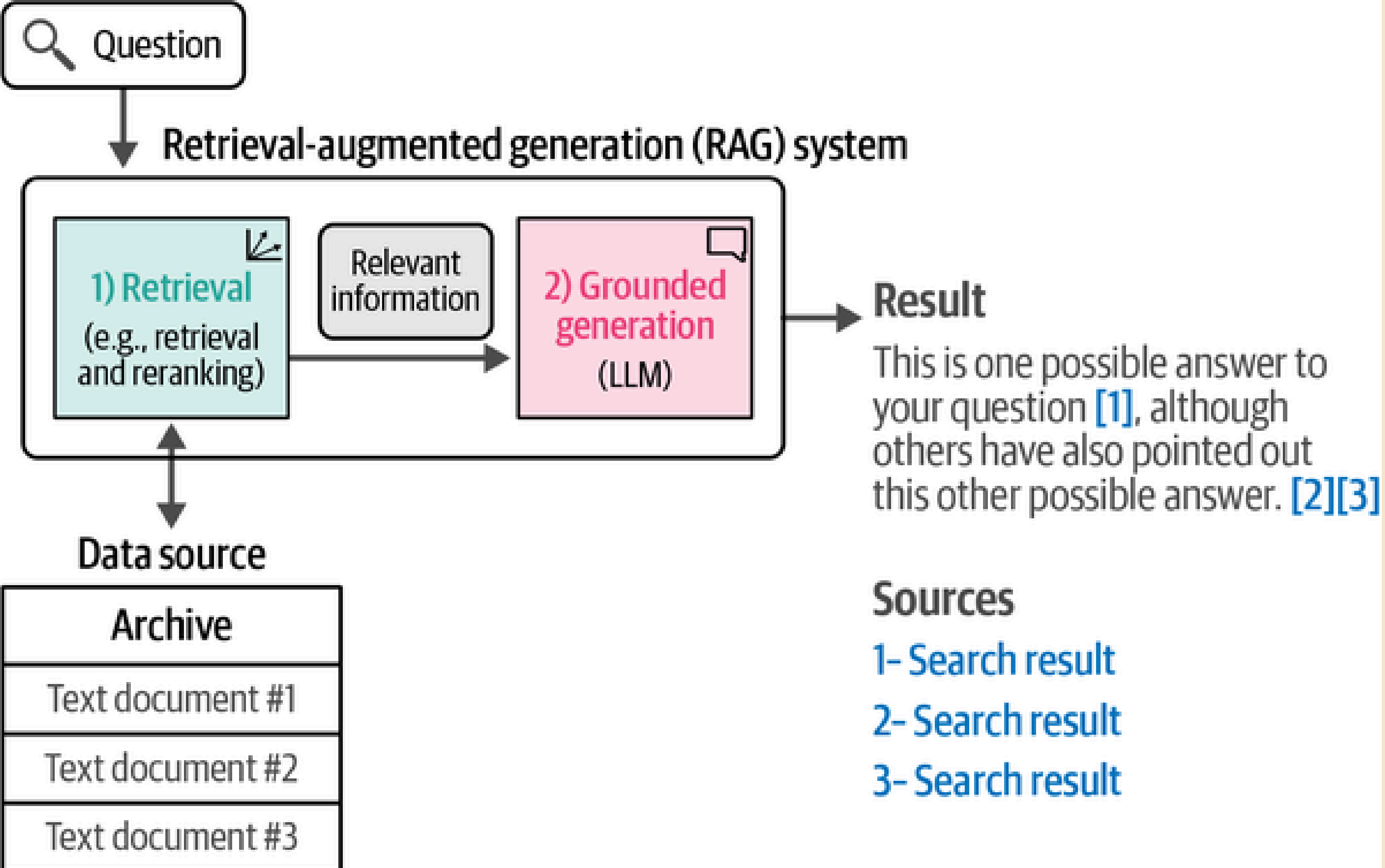


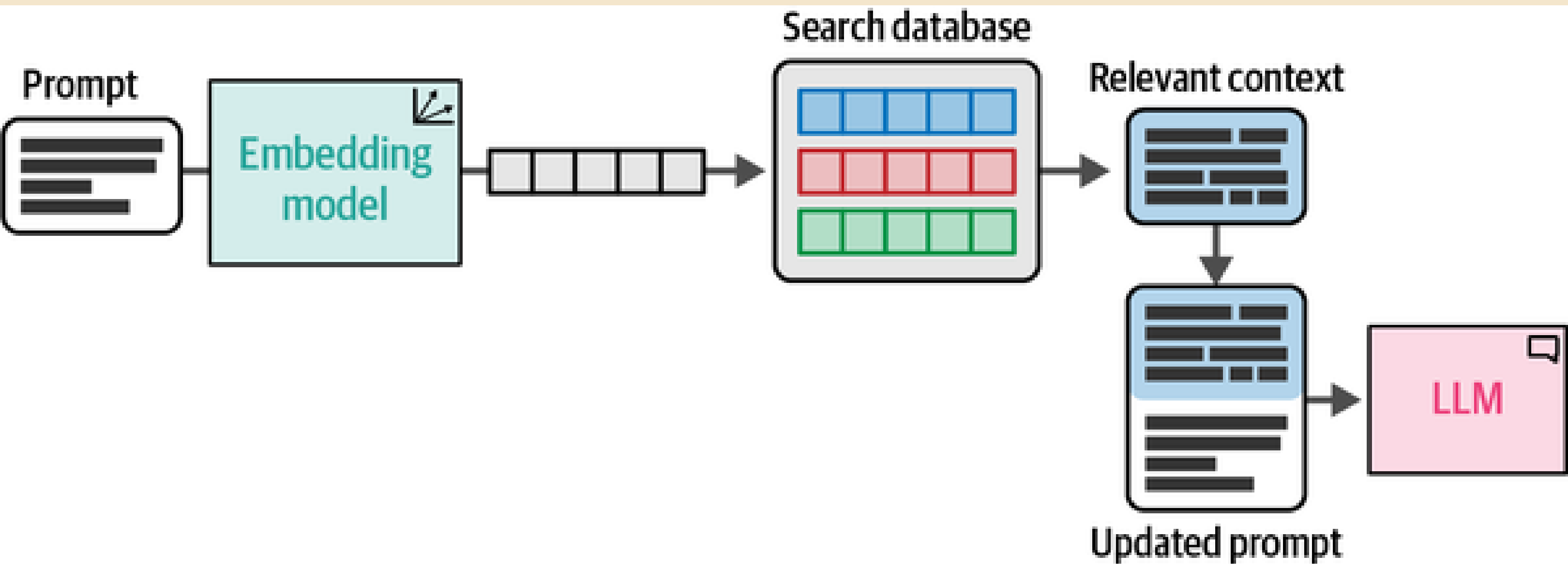


RAG（檢索增強生成）

Retrieval-augmented generation (RAG) system







下一章

多模態大型語言模型

